1 a. What are S & A

$S = \{(x, y, c) : (x, y)$ is a non-darkwall, $c \in \{r, g, b\}\}$
∴ $|S| = 81 - 13 = 68 \cdot c = 204$

A = is the set of all possible actoins → {Up, down, left, Right}
∴ $|A| = 4$

b. We have $204 \cdot 4 = 816$ number of s, a pairs & we have 3 outcomes per pair. Therefore We have about 2,450 non zero rows but because of wall merges it's closer to 2,400.

2. a. $G_T = R_{t+1} + \gamma R_{t+2} + \gamma^3 R_{t+3} \ldots \rightarrow G_T = R_{t+1} + \gamma G_{t+1}$
$G_5 = R_6 + \gamma R_7$ but T stops at 5 there is no R_6 so $G_5 = 0$
$G_4 = 4 + 0.5 \cdot G_5 = 4$; $G_3 = R_4 + 0.5 \cdot R_5 = 2 + 0.5 \cdot 4 = 4$
$G_2 = R_3 + 0.5 R_4 + 0.5^2 + R_5 \Rightarrow 0 + (0.5 \cdot 2) + (.25 \cdot 4) = 2$
$G_1 = R_2 + 2 \rightarrow$ from $G_2 \Rightarrow 1 + .5 \cdot 2 = 2$
$G_0 = R_1 + 0.5 \cdot G_1 = 6$

b. $G_0 = \gamma^{10} 100$ for option 1
Option 2 is a geometric series ∴ we can call $S = 1 + y \ldots + y^n$ then we multiply both sides by $y \rightarrow S\gamma = \gamma + \gamma^2 + \ldots + \gamma^n$ now $S - \gamma S = 1$ because everythig cancels out except the 1. now factor out $S \rightarrow S(1-\gamma) = 1$
& ∴ $S = 1/(1-\gamma)$, let's check with $\gamma = 0.5 \rightarrow = 2$
$\gamma^{10} 100 = 1/(1-\gamma)$. Using desmos I got $\gamma = 0.715$ & $\gamma = 0.988$.

3. The issue is that the number of time steps is not taken into cosideration when calculating a return. The agent would get the same return per episode. If we change $\gamma = 0.9$ we can use a discount and the escaping step gives us $G_0 = \gamma^{T-1}$, allowing a reward at timestep 4 to be valued more than at timestep 20.

4. $G_T = \sum_{k=0}^{\infty} \gamma^k (R_{t+k+1} + c) \Rightarrow G_T = \underbrace{\sum_{k=0}^{\infty} \gamma^k R_{t+k+1}}_{\text{original } G_T} + \sum_{k=0}^{\infty} \gamma^k c$

for this piece $\sum_{k=0}^{\infty} \gamma^k c \Rightarrow c \sum_{k=0}^{\infty} \gamma^k \Rightarrow \boxed{c \cdot 1/(1-\gamma) = V_c}$

$\therefore G_T' = G_T + V_c$ - V_c is the same for every policy so the relative values dont change, therefore only the intervals for the reward matter not the sign.

5. a) $\gamma = 0.9$, there is no reward for any of the 4 actions. π^* shows us optimal action is west or north, $\therefore v^* = 1/2(17.8) + 1/2(17.8) \approx 17.8$. We split the two optimal action respectvly

b) each action has $1/4$ chance being taken so $\pi(a|s) = 1/4$. In addition the grip world is deterministic so $p(s', r | s, a) = 1$. $\therefore$ const

$$V_\pi(s) = \sum_a \pi(a|s)[r + \gamma \cdot V_\pi(s)]$$

$\therefore 1/4[0 + (0.9 \cdot 2.3) + (0.9 \cdot 0.7) + (0.9 \cdot 0.4) + (0.9 \cdot -0.4)$

$= 1/4 \cdot 0.9 \cdot 3.0 = \boxed{0.675 \approx .7}$ and matches the table closley enogh

6. $v(high) = \pi(search|high)[a(r_{search} + \gamma v(high) + (1-a)[r_search + \gamma v(low)] + \pi(wait|high)[r_wait + \gamma v(high)]$

now for low

$v(low) = \pi(search|low)[\beta(r_search + \gamma v(low) + (1-\beta)[-3 + \gamma v(high)]$
$+ \pi(wait|low)[r_wait + \gamma v(low)] + \pi(recharge|low)[\gamma v(high)]$

b. $v(high) = 1 \cdot [0.75(3 + 0.9 v(high)] + 0.25(3 + 0.9 v(low)$
$= [.225 + .675 v(high) + .75 + .225 v(low)$
$= 3 + .675 v(high) + .225 v(low)$
$0.325 v(high) - 0.225 v(low) = 3$

$v(low) = 0.2(1 + 0.9 v(low)) + 0.8(0.9 v(high)$
multiply out
$v(low) = 0.2 + .18 v(low) + 0.72 v(high)$
$= -0.72 v(high) + 0.82 v(low) = 0.2$
now let's use substitution to solve
$v(high) = 3 + 0.225 v(low) / 0.325$

$-0.72[3 + 0.225 v(low) / 0.325] + 0.82 v(low) = 0.2$
$= 2.2154(3 + 0.225 v(low)$
$= -6.646 - 0.498 v(low) + 0.82 v(low) = 0.2$
$= 0.3215 v(low) = 6.846$ $\boxed{v(low) = 21.29}$

$v(high) = (3 + 0.225 \times 21.29) / 0.325) = (3 + 4.79) / 0.325 = 23.97$
$\boxed{v(high) = 23.97}$